

Typing Rules and Pi Injectivity for Dependent Type Theory with $\mathbf{U} : \mathbf{U}$

1 Judgments

We work with three mutually defined judgment forms:

$\Gamma \text{ ctx}$	Γ is a well-formed context
$\Gamma \vdash M : A$	M has type A in context Γ
$\Gamma \vdash M = N : A$	M and N are definitionally equal at type A

Contexts are snoc-lists: $\Gamma ::= \cdot \mid \Gamma, x:A$.

2 Well-formed contexts

$$\frac{}{\cdot \text{ ctx}} \qquad \frac{\Gamma \vdash A : \mathbf{U}}{\Gamma, x:A \text{ ctx}}$$

3 Typing rules

$$\begin{array}{c}
\text{VAR} \\
\frac{\Gamma \text{ ctx} \quad (x:A) \in \Gamma}{\Gamma \vdash x : A}
\end{array}
\qquad
\begin{array}{c}
\text{CONV} \\
\frac{\Gamma \vdash M : A \quad \Gamma \vdash A = B : \mathbf{U} \quad \Gamma \vdash B : \mathbf{U}}{\Gamma \vdash M : B}
\end{array}
\qquad
\begin{array}{c}
\mathbf{U} \\
\frac{\Gamma \text{ ctx}}{\Gamma \vdash \mathbf{U} : \mathbf{U}}
\end{array}$$

$$\begin{array}{c}
\text{PI} \\
\frac{\Gamma \vdash A : \mathbf{U} \quad \Gamma, x:A \vdash B : \mathbf{U}}{\Gamma \vdash \Pi(x:A)B : \mathbf{U}}
\end{array}
\qquad
\begin{array}{c}
\text{LAM} \\
\frac{\Gamma \vdash A : \mathbf{U} \quad \Gamma, x:A \vdash B : \mathbf{U} \quad \Gamma, x:A \vdash M : B}{\Gamma \vdash \lambda(x:A)M : \Pi(x:A)B}
\end{array}$$

$$\begin{array}{c}
\text{APP} \\
\frac{\Gamma \vdash A : \mathbf{U} \quad \Gamma, x:A \vdash B : \mathbf{U} \quad \Gamma \vdash f : \Pi(x:A)B \quad \Gamma \vdash a : A}{\Gamma \vdash f a : B[a]}
\end{array}$$

4 Conversion rules

$$\begin{array}{c}
\text{REFL} \\
\frac{\Gamma \vdash M : A}{\Gamma \vdash M = M : A} \\
\\
\text{SYM} \\
\frac{\Gamma \vdash M = N : A}{\Gamma \vdash N = M : A} \\
\\
\text{TRANS} \\
\frac{\Gamma \vdash M = N : A \quad \Gamma \vdash N = P : A}{\Gamma \vdash M = P : A} \\
\\
\text{CONV-EQ} \\
\frac{\Gamma \vdash M = N : A \quad \Gamma \vdash A = B : \mathbb{U} \quad \Gamma \vdash B : \mathbb{U}}{\Gamma \vdash M = N : B} \\
\\
\text{BETA} \\
\frac{\Gamma \vdash A : \mathbb{U} \quad \Gamma, x:A \vdash B : \mathbb{U} \quad \Gamma, x:A \vdash M : B \quad \Gamma \vdash a : A}{\Gamma \vdash (\lambda(x:A)M) a = M[a] : B[a]} \\
\\
\text{PI-CONG} \\
\frac{\Gamma \vdash A = A' : \mathbb{U} \quad \Gamma, x:A \vdash B = B' : \mathbb{U}}{\Gamma \vdash \Pi(x:A)B = \Pi(x:A')B' : \mathbb{U}} \\
\\
\text{FUNEXT} \\
\frac{\Gamma \vdash A : \mathbb{U} \quad \Gamma, x:A \vdash f^+ x = g^+ x : B \quad \Gamma \vdash f : \Pi(x:A)B \quad \Gamma \vdash g : \Pi(x:A)B}{\Gamma \vdash f = g : \Pi(x:A)B} \\
\\
\text{APP-FUN} \\
\frac{\Gamma \vdash A : \mathbb{U} \quad \Gamma, x:A \vdash B : \mathbb{U} \quad \Gamma \vdash f = f' : \Pi(x:A)B \quad \Gamma \vdash a : A}{\Gamma \vdash f a = f' a : B[a]} \\
\\
\text{APP-ARG} \\
\frac{\Gamma \vdash A : \mathbb{U} \quad \Gamma, x:A \vdash B : \mathbb{U} \quad \Gamma \vdash f : \Pi(x:A)B \quad \Gamma \vdash a = a' : A}{\Gamma \vdash f a = f a' : B[a]}
\end{array}$$

5 Head reduction

We define head reduction $M \rightsquigarrow N$ as the compatible closure of β -reduction at the head:

$$\begin{array}{c}
\text{HR-BETA} \\
\frac{}{(\lambda(x:A)M) a \rightsquigarrow M[a]} \\
\\
\text{HR-APP} \\
\frac{M \rightsquigarrow M'}{M a \rightsquigarrow M' a}
\end{array}$$

6 Main result: Pi injectivity (head reduction)

Theorem 1 (Corollary 6, part 1; Coquand–Huber 2018, p. 661). *If $\Gamma \vdash A_0 = \Pi(x:B_1)F_1 : \mathbb{U}$, then there exist B_0, F_0 such that*

$$A_0 \rightsquigarrow^* \Pi(x:B_0)F_0.$$

Proof outline. The proof proceeds by domain-theoretic adequacy.

1. From $\Gamma \vdash A_0 = \Pi B_1 F_1 : \mathbb{U}$, extract $\Gamma \vdash A_0 : \mathbb{U}$ by presupposition (SUBSTITUTIONLEMMA).
2. Apply the soundness theorem (CONVSOUND') at the *bottom environment* $\rho_\perp$ (mapping every variable to $\perp$) to obtain a bidirectional evaluation transfer: for every finite approximation u ,

$$\text{EvalRel}(\Pi B_1 F_1, \rho_\perp, u) \implies \text{EvalRel}(A_0, \rho_\perp, u).$$

3. Instantiate with $u = \text{PiCode}(\perp, \text{nil})$, which is coherent because $\text{CoherentFunTail}(\text{nil}) = \top$. The evaluation $\text{EvalRel}(\Pi B_1 F_1, \rho_\perp, \text{PiCode}(\perp, \text{nil}))$ is constructible trivially (domain $\perp$, empty function graph, vacuous selection body). Transfer gives $\text{EvalRel}(A_0, \rho_\perp, \text{PiCode}(\perp, \text{nil}))$.
4. Apply the bundled adequacy theorem (`ADEQUACYSUB2`) with the identity substitution at $\rho_\perp$ to obtain

$$\text{Val2 } \Gamma \ A_0 \ \text{U } (\text{PiCode}(\perp, \text{nil})) \ \text{UCode}.$$

5. By definition, Val2 at $(\text{PiCode}(\perp, \text{nil}), \text{UCode})$ unfolds to $\text{ValTyPi2 } \Gamma \ A_0 \ \perp \ \text{nil}$, which is a Σ -type whose third component is a contextual reduction $\text{Red } \Gamma \ A_0 \ (\Pi B_0 F_0) \ \text{U}$. Extract the underlying head reduction $A_0 \rightsquigarrow^* \Pi B_0 F_0$.

The entire proof is formalised in Agda with `--without-K` `--exact-split`, no `--type-in-type`, and no postulates (in the files on which `PiInjectivity.agda` depends). $\square$

References

- [1] T. Coquand and S. Huber, “An adequacy theorem for dependent type theory,” *Archive for Mathematical Logic*, vol. 57, pp. 649–676, 2018.